FULL STACK

LAB SHEET 5

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SEC – B, ROLL – 16

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Express Lab</title>
  <link rel="stylesheet" href="/style.css">
</head>
<body>
  <h1>Express.js Lab</h1>
  <p>Express Lab Running</p>
  <p>Library Management Mini API is ready.</p>
</body>
</html>
```

Style.css

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: "Segoe UI", Arial, sans-serif;
```

```
    min-height: 100vh;
    background: linear-gradient(135deg, #eef2ff, #f8fafc);
    color: #1e293b;
    display: flex;
    justify-content: center;
    align-items: center;
    padding: 30px;
}

.container {
    width: 100%;
    max-width: 850px;
    background: #ffffff;
    padding: 45px;
    border-radius: 20px;
    box-shadow: 0 15px 45px rgba(15, 23, 42, 0.12);
}

h1 {
    font-size: 36px;
    color: #1e3a8a;
    margin-bottom: 12px;
}

h2 {
    font-size: 22px;
    color: #334155;
    margin: 30px 0 15px;
}

p {
    color: #64748b;
    font-size: 16px;
```

```
    line-height: 1.7;
    margin-bottom: 20px;
}

.card {
    background: #f8fafc;
    border: 1px solid #e2e8f0;
    border-radius: 14px;
    padding: 22px;
    margin-top: 20px;
    transition: 0.25s ease;
}

.card:hover {
    transform: translateY(-3px);
    box-shadow: 0 8px 25px rgba(15, 23, 42, 0.08);
}

ul {
    list-style: none;
    margin-top: 15px;
}

li {
    padding: 12px 15px;
    margin-bottom: 8px;
    background: #ffffff;
    border-left: 4px solid #2563eb;
    border-radius: 8px;
    color: #475569;
}

a {
```

```
    color: #2563eb;
    text-decoration: none;
    font-weight: 600;
}

a:hover {
    text-decoration: underline;
}

button {
    border: none;
    background: #2563eb;
    color: white;
    padding: 12px 22px;
    border-radius: 8px;
    font-size: 15px;
    font-weight: 600;
    cursor: pointer;
    transition: 0.25s ease;
}

button:hover {
    background: #1d4ed8;
    transform: translateY(-2px);
}

input,
textarea,
select {
    width: 100%;
    padding: 12px 14px;
    margin-top: 8px;
    margin-bottom: 18px;
```

```
    border: 1px solid #cbd5e1;
    border-radius: 8px;
    font-size: 15px;
    outline: none;
    transition: 0.2s ease;
}

input:focus,
textarea:focus,
select:focus {
    border-color: #2563eb;
    box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.12);
}

.status {
    display: inline-block;
    padding: 7px 14px;
    border-radius: 20px;
    background: #dcfce7;
    color: #166534;
    font-size: 13px;
    font-weight: 600;
}

footer {
    margin-top: 35px;
    padding-top: 20px;
    border-top: 1px solid #e2e8f0;
    text-align: center;
    color: #94a3b8;
    font-size: 14px;
}
```

```
@media (max-width: 600px) {
  body {
    padding: 15px;
  }

  .container {
    padding: 25px;
    border-radius: 14px;
  }

  h1 {
    font-size: 28px;
  }

  h2 {
    font-size: 20px;
  }
}
```

Index.js

```
require('dotenv').config();
const express = require('express');
const path = require('path');

const studentsRouter = require('./routes/students');
const booksRouter = require('./routes/books');
const membersRouter = require('./routes/members');
const tasksRouter = require('./routes/tasks');

const app = express();
const PORT = process.env.PORT || 4000;

// Part D: global request logger
```

```
app.use((req, res, next) => {
  console.log(`[${new Date().toISOString()}] ${req.method} ${req.originalUrl}`);
  next();
});

// Part E: JSON and form parsing
app.use(express.json());
app.use(express.urlencoded({ extended: true }));

// Part I: static files
app.use(express.static(path.join(__dirname, 'public')));

// Part D: fake API key for /admin routes
app.use('/admin', (req, res, next) => {
  if (!req.headers['x-api-key']) {
    return res.status(403).json({ error: 'Forbidden' });
  }
  next();
});

// Part A
app.get('/', (req, res) => res.send('Express Lab Running'));

// Part B
app.get('/about', (req, res) => {
  res.json({ name: 'Pallav Pankaj', rollNumber: '40' });
});

app.get('/courses', (req, res) => {
  res.json(['Full Stack Web Development', 'Data Structures', 'Computer Vision']);
});

app.post('/echo', (req, res) => res.json(req.body));
```

```
// Part C
app.get('/search', (req, res) => {
  res.json({ name: req.query.name, age: req.query.age });
});

app.get('/products/:category/:id', (req, res) => {
  res.json({ category: req.params.category, id: req.params.id });
});

// Part D
app.get('/admin/dashboard', (req, res) => {
  res.json({ message: 'Admin dashboard data', authenticated: true });
});

// Part E
app.post('/register', (req, res) => {
  const { name, email, password } = req.body;
  res.json({ message: `Registration successful for ${name}`, email, passwordReceived: Boolean(password) });
});

app.post('/contact', (req, res) => {
  console.log('Contact form:', req.body);
  res.send(`Thank you for contacting us, ${req.body.name || 'user'}!`);
});

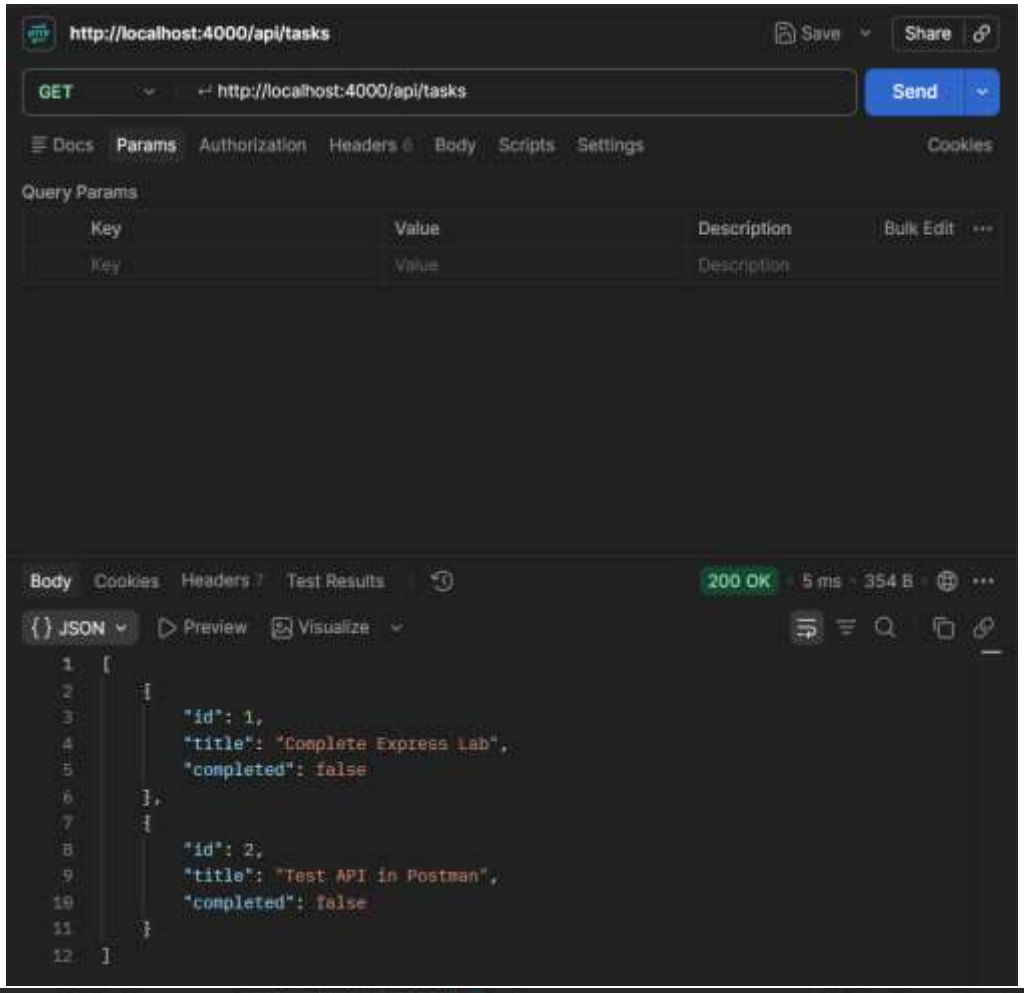
// Part F/J routers
app.use('/api/students', studentsRouter);
app.use('/api/books', booksRouter);
app.use('/api/members', membersRouter);
app.use('/api/tasks', tasksRouter);

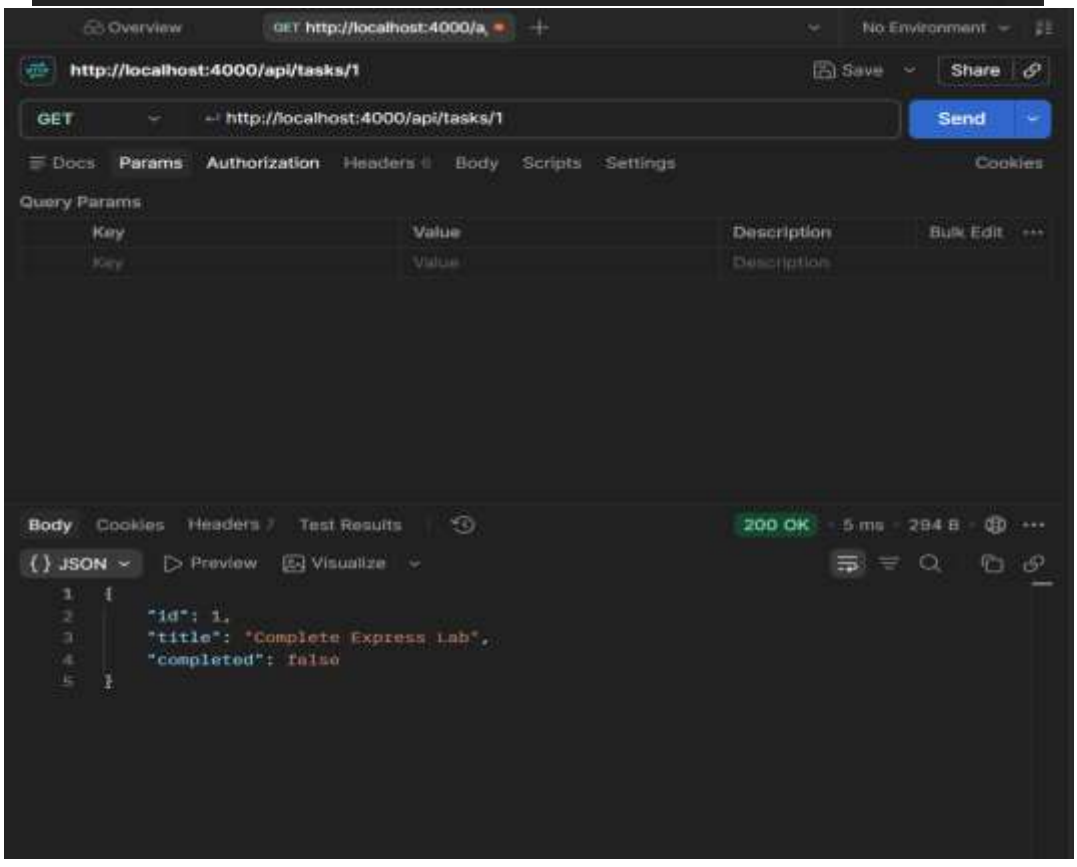
// Part H: 404 handler
```

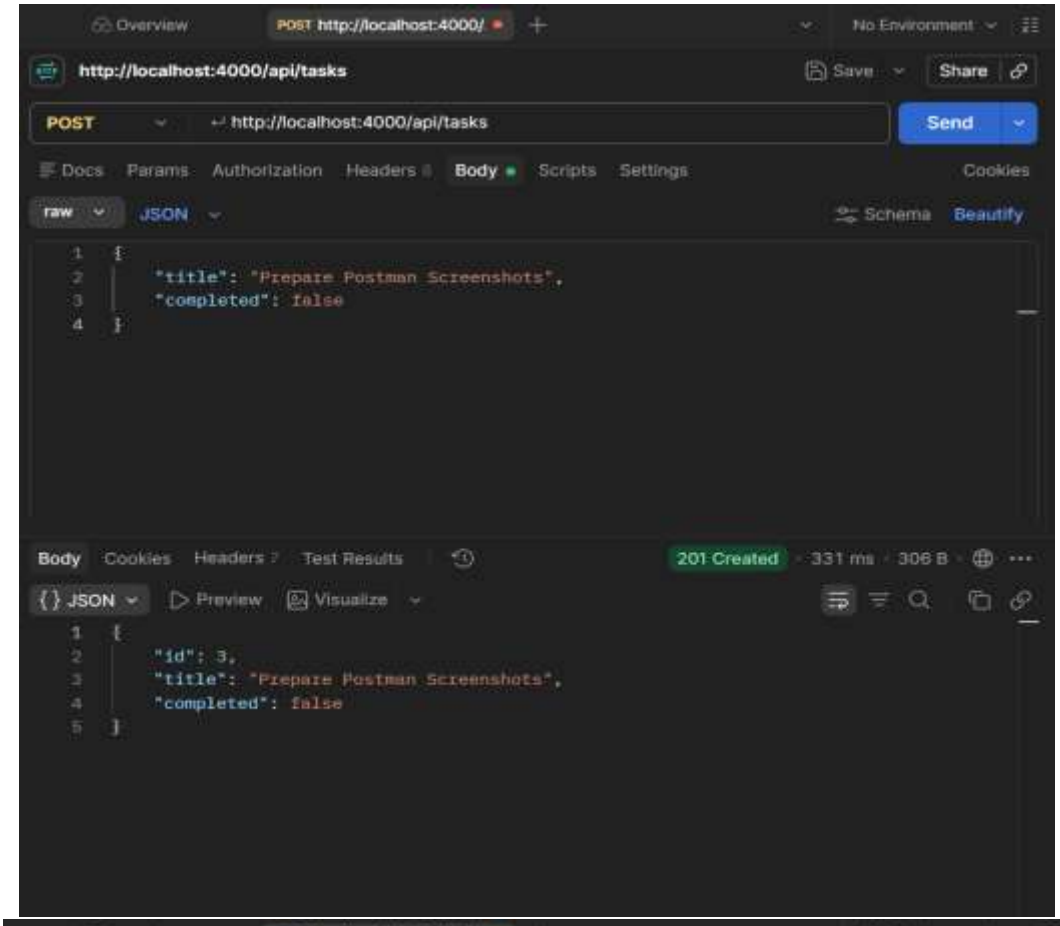
```
app.use((req, res, next) => {
  const err = new Error('Route not found');
  err.status = 404;
  next(err);
});

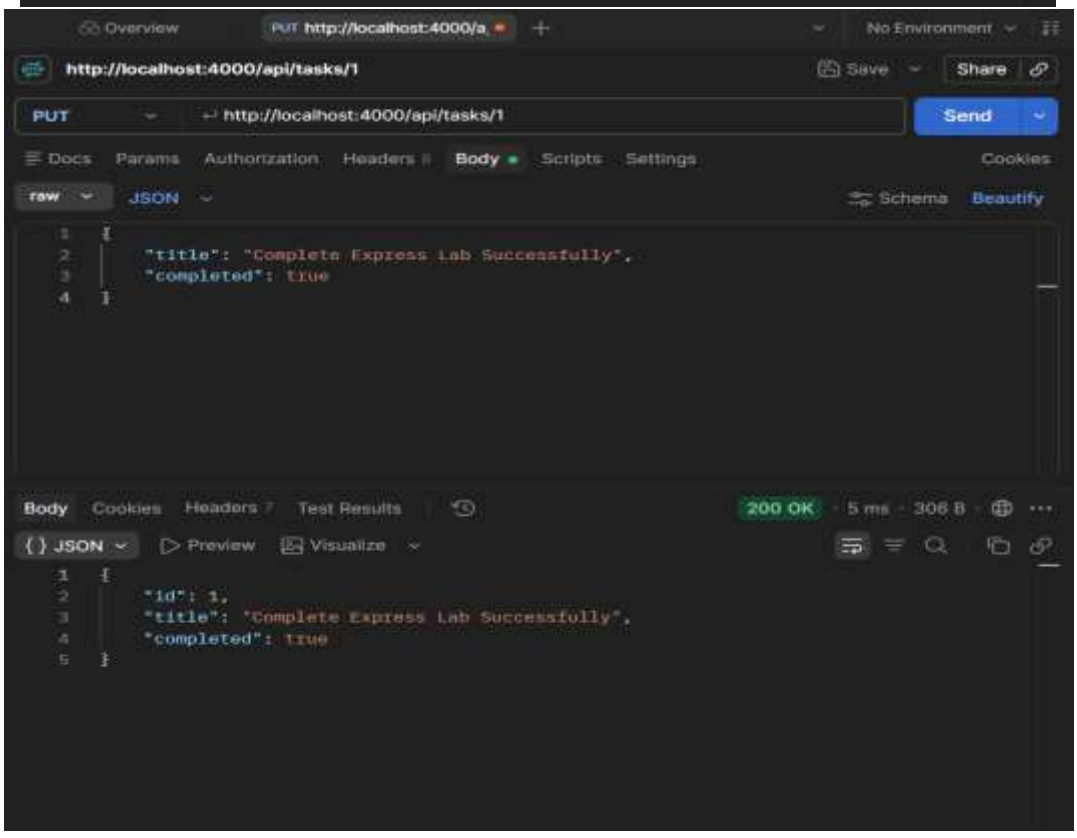
// Part H: centralized error handler
app.use((err, req, res, next) => {
  console.error(err.message);
  const status = err.status || 500;
  res.status(status).json({ error: err.message || 'Internal server error' });
});

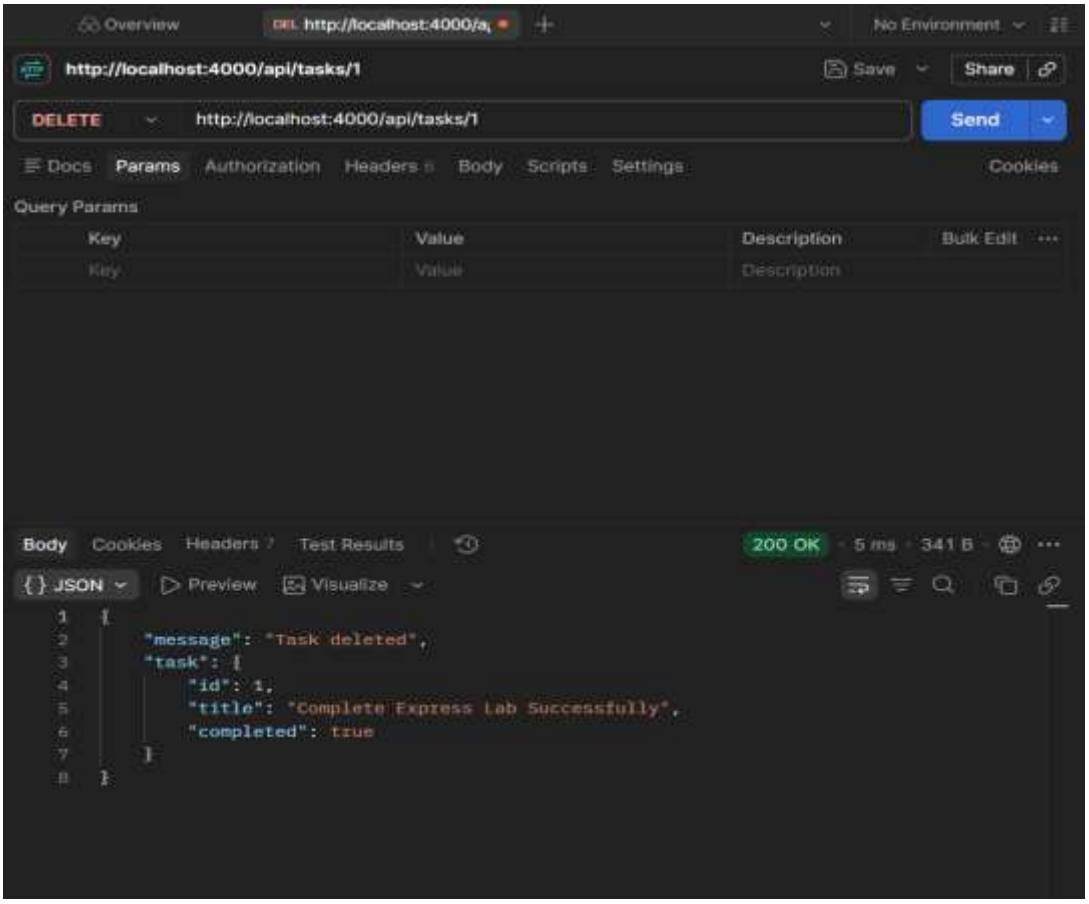
app.listen(PORT, () => console.log(`Server running at http://localhost:${PORT}`));
```


http://localhost:4000/api/tasks
Save
Share
GET
http://localhost:4000/api/tasks
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Key
Value
Description
Body
Cookies
Headers 7
Test Results
200 OK
5 ms
354 B
{} JSON
Preview
Visualize
[
{
"id": 1,
"title": "Complete Express Lab",
"completed": false
},
{
"id": 2,
"title": "Test API in Postman",
"completed": false
}
]


Overview
GET http://localhost:4000/a
No Environment
http://localhost:4000/api/tasks/1
Save
Share
GET
http://localhost:4000/api/tasks/1
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Key
Value
Description
Body
Cookies
Headers 7
Test Results
200 OK
5 ms
294 B
{} JSON
Preview
Visualize
{
"id": 1,
"title": "Complete Express Lab",
"completed": false
}


Overview
POST http://localhost:4000/
No Environment
http://localhost:4000/api/tasks
Save
Share
POST
http://localhost:4000/api/tasks
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"title": "Prepare Postman Screenshots",
"completed": false
}
Body
Cookies
Headers 7
Test Results
201 Created
331 ms
306 B
{ } JSON
Preview
Visualize
{
"id": 3,
"title": "Prepare Postman Screenshots",
"completed": false
}


Overview
PUT http://localhost:4000/a
No Environment
http://localhost:4000/api/tasks/1
Save
Share
PUT
http://localhost:4000/api/tasks/1
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"title": "Complete Express Lab Successfully",
"completed": true
}
Body
Cookies
Headers 7
Test Results
200 OK
5 ms
306 B
{ } JSON
Preview
Visualize
{
"id": 1,
"title": "Complete Express Lab Successfully",
"completed": true
}


Overview
DEL http://localhost:4000/a
No Environment
http://localhost:4000/api/tasks/1
Save
Share
DELETE
http://localhost:4000/api/tasks/1
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Key
Value
Description
Body
Cookies
Headers 7
Test Results
200 OK
5 ms
341 B
{ } JSON
Preview
Visualize
{
"message": "Task deleted",
"task": {
"id": 1,
"title": "Complete Express Lab Successfully",
"completed": true
}
}